

Ivonne Mariela García Martínez, Ph.D.

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🎯 SNII Candidate, SECIHTI

Education

- 2016 – 2020 📖 **Ph.D. Atmospheric and Environmental Sciences.** The University of Edinburgh. Edinburgh, UK.
Thesis title: *Mechanisms of climate variation in the broader Mexican-United States region: the role of atmospheric circulation and anthropogenic forcing.*
- 2012 – 2015 📖 **M.Sc. Physical Oceanography.** Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, Mexico.
Thesis title: *Ocean-atmosphere variability related with the midsummer drought in the CFSR reanalysis.*
- 2007 – 2011 📖 **B.Sc. Atmospheric Sciences.** Universidad Veracruzana. Xalapa, Mexico.
Graduated by high GPA.

Research Publications

Peer-reviewed articles

- 1 **I. M. García-Martínez**, J. Sheinbaum, L. Zavala Sansón, J. Jouanno, and E. Pallàs-Sanz, “Upper-ocean response to strong wind forcing in a Loop Current Eddy,” *Journal of Physical Oceanography*, vol. 55, no. 12, pp. 2401–2420, 2025. 📄 DOI: 10.1175/JPO-D-25-0022.1.
- 2 L. Zavala Sansón, **I. M. García-Martínez**, and J. Sheinbaum, “Power input generated by a linear wind field on circular and elliptical ocean vortices,” *Journal of Physical Oceanography*, vol. 55, no. 9, pp. 1493–1506, 2025. 📄 DOI: 10.1175/JPO-D-24-0114.1.
- 3 L. Zavala Sansón, **I. M. García-Martínez**, and J. Sheinbaum Pardo, “Nonlinear surface Ekman effects on cyclonic and anticyclonic vortices,” *Journal of Fluid Mechanics*, vol. 971, A35, 2023. 📄 DOI: 10.1017/jfm.2023.688.
- 4 **I. M. García-Martínez** and M. A. Bolasina, “Identifying the evolving human imprint on heat wave trends over the United States and Mexico,” *Environmental Research Letters*, vol. 16, no. 9, p. 094 039, 2021. 📄 DOI: 10.1088/1748-9326/ac1edb.
- 5 **I. M. García-Martínez**, M. A. Bolasina, and S. Undorf, “Strong large-scale climate response to North American sulphate aerosols in CESM,” *Environmental Research Letters*, vol. 15, no. 11, p. 114 051, 2020. 📄 DOI: 10.1088/1748-9326/abbe45.
- 6 **I. M. García-Martínez** and M. A. Bolasina, “Sub-monthly evolution of the Caribbean Low-Level Jet and its relationship with regional precipitation and atmospheric circulation,” *Climate Dynamics*, vol. 54, pp. 4423–4440, 2020. 📄 DOI: 10.1007/s00382-020-05237-y.

Book chapters

- 1 A. Tejeda Martínez, D. R. Fitzjarrald, M. Pérez Méndez, *et al.*, “Ciclos diurnos y circulación local (Diurnal cycles and local circulation),” in *Un laboratorio natural del clima: la interacción costa-montaña en el centro del Golfo de México*, A. Tejeda Martínez, M. Pérez Méndez, and I. R. Méndez Pérez, Eds., Xalapa, México: Universidad Veracruzana, 2020, pp. 46–91.
- 2 A. Tejeda Martínez, **I. M. Garcia-Martínez**, G. Gómez-Azpeitia, I. R. Méndez-Pérez, and P. J. González Viveros, “Procedimientos simples para la estimación de radiación Solar en México (Simple procedures for the estimation of solar radiation in Mexico),” in *Prontuario solar de México*, G. Gómez-Azpeitia and A. Tejeda Martínez, Eds., Colima, México: Universidad de Colima and Universidad Veracruzana, 2015, pp. 103–112.
- 3 A. Tejeda Martínez, **I. M. Garcia-Martínez**, I. R. Méndez-Pérez, U. Miranda-Miranda, and J. V. López-Méndez, “Radiación solar en México (Solar radiation in Mexico),” in *Prontuario solar de México*, G. Gómez-Azpeitia and A. Tejeda Martínez, Eds., Colima, México: Universidad de Colima and Universidad Veracruzana, 2015, pp. 51–68.

Manuscripts in submission and in preparation

- 1 **I. M. Garcia-Martínez** and M. A. Bollasina, “Projected compound hot and dry events over Mexico in HighResMIP GCMs,” In preparation. To be submitted to Weather and Climate Extremes.
- 2 **I. M. Garcia-Martínez**, J. Jouanno, J. Rodríguez, *et al.*, “Dynamical response of a Loop Current Eddy to strong wind forcing in high-resolution coupled ocean-atmosphere simulations,” In preparation. To be submitted to Journal of Geophysical Research: Oceans.

Manuscript reviews

- Climate Dynamics
- International Journal of Climatology
- GEOS

Awarded scholarships

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| 2016 – 2020 | <ul style="list-style-type: none"> ■ PhD Scholarship from the National Council for Science and Technology (CONA-CyT, Mexico). ■ Complementary PhD scholarship from the University of Edinburgh, UK. |
| 2012 – 2014 | <ul style="list-style-type: none"> ■ MSc Scholarship from the National Council for Science and Technology (CONA-CyT, Mexico). |

Academic employment

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| 2024 – | <ul style="list-style-type: none"> ■ Research Associate. Department of Physical Oceanography. Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México. |
| 2021 – 2023 | <ul style="list-style-type: none"> ■ Postdoctoral Researcher, Department of Physical Oceanography. Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México. |
| 2015 – 2016 | <ul style="list-style-type: none"> ■ Junior Researcher (M.Sc. level), Department of Physical Oceanography. Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México. |
| 2011 – 2012 | <ul style="list-style-type: none"> ■ Research Assistant, Applied Climatology Group. Universidad Veracruzana. Xalapa, México. |

Research visits

- 2022 –  **Remote visiting researcher, School of GeoSciences.** The University of Edinburgh. Edinburgh, UK.
- 2025  **Laboratoire d'Etudes en Géophysique et Oceanographie Spatiales, LEGOS.** Toulouse, France.

Teaching experience

- 2022, 2025  **Climatology and Climate Change.** Postgraduate program in Physical Oceanography. Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE). Ensenada, México. Lecturer.
- 2016 – 2018  **Earth Modelling and Prediction.** School of GeoSciences. The University of Edinburgh. Edinburgh, UK. Teaching assistant.

Invited talks

- 2022  **Weather of 2021 in the context of the long-term climate change.** "Understanding the Weather of 2021" virtual event of the Royal Meteorological Society of the UK. 26 February 2022. <https://www.rmets.org/event/virtual-understanding-weather-2021>
-  **State of the climate in 2021 under the context of climate change.** Universidad Veracruzana. Invited talk as part of the University's celebrations on the World Meteorological Day. 23 March 2022.
- 2021  **Heat wave trends in Mexico and the United States: contribution of anthropogenic aerosols, greenhouse gases and internal variability.** Universidad Veracruzana. Invited seminar in the 45th Anniversary of the School of Atmospheric Sciences. 6 May 2021.

Related professional and technical skills

- Languages  Strong reading, writing and speaking skills in English. Native Spanish and basic Portuguese.
- Coding  Python, Matlab, shell, NCO, CDO, GrADS, and high performance computing
- Modelling  WRF (atmospheric model), NEMO and CROCO (ocean models), OASIS (coupler)
- Version control  Git
- Editing  L^AT_EX

Additional training

- 2018  **ICTP Summer School** on Theory, Mechanisms and Hierarchical Modelling of Climate Dynamics: Multiple Equilibria in the Climate System. Trieste, Italy, 25-30 June 2018. At International Centre for Theoretical Physics.
-  **WCRP Grand Challenge** on Clouds, Circulation and Climate Sensitivity: 2nd meeting on Monsoons and Tropical Rain Belts. Trieste, Italy, 2 - 5 July 2018. At International Centre for Theoretical Physics.
- 2017  **GOTHAM Summer School** on Global Teleconnections in the Earth's Climate System: Processes, Modelling and Advanced Analysis Methods. Potsdam, Germany, 18-22 September 2017. At Potsdam Institute for Climate Impact Research.

Additional training (continued)

- 2014  **XXII Alpine Summer School** on Dynamics, Stochastics and Predictability of the Climate System. Valsavarenche, Valle d'Aosta, Italy, June 9-18, 2014.

Participation in conferences and other academic activities

-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 26 October to November 1, 2025. Observational diagnosis of the footprint depth of marine heat waves in the Caribbean Sea. Romero, C., Cetina P and **García-Martínez, I. M.** Oral presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 26 October to November 1, 2025. Modulation of vertical velocity in an anticyclonic vortex due to the passage of Nortes in the Gulf of Mexico. Jiménez, K., **García-Martínez, I. M.** and Others. Poster.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 27 October to November 2, 2024. Compound hot and dry events in Mexico: current scenarios and projections to 2050 based on models from the HIGHRESMIP project. **García-Martínez, I. M.** and M. Bollasina. Oral presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 27 October to November 2, 2024. Coupled and non-coupled numerical simulations of the evolution of a Loop Current Eddy during a Norte (strong-wind) event. J. Sheinbaum, **García-Martínez, I. M.**, J. Jouanno and Others. Oral presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 29 October to November 3, 2023. Towards a coupled ocean-atmosphere modelling system of the Gulf of Mexico: eddy-wind interactions. **García-Martínez, I. M.** and J. Sheinbaum. Oral presentation.
-  European Geosciences Union (**EGU**) General Assembly. Vienna, Austria, April 23-28, 2023. Current feedback in an anticyclonic eddy during the passage of cold frontal systems over the Gulf of Mexico. **García-Martínez I. M.** and J. Sheinbaum. Poster.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 30 October to November 4, 2022. Influence of strong winds in an anticyclonic eddy in the Gulf of Mexico. **García-Martínez, I. M.** and J. Sheinbaum. Oral presentation.
-  Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, 30 October to November 4, 2022. Heat wave trends in Mexico and the United States: contribution from anthropogenic aerosols, greenhouse gases and internal variability. **García-Martínez, I. M.** and M. Bollasina. Oral presentation.
-  World Climate Research Program (**WCRP**) virtual workshop on attribution of multiannual to decadal changes in the climate system. September 22-24, 2021. Identifying the evolving human imprint on heat wave trends over Mexico and the United States. **García-Martínez, I. M.** and M. Bollasina. Oral presentation.

Participation in conferences and other academic activities (continued)

- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, November 2-8, 2019. Sub-monthly evolution of the Caribbean low-level jet and its relationship with precipitation in Mexico. **García-Martínez, I. M.** and M. Bollasina. Oral presentation.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, November 2-8, 2019. Anthropogenic aerosols as drivers of Mexican and Central American climate. **García-Martínez, I. M.** and M. Bollasina. Poster.
- European Geosciences Union (**EGU**) General Assembly. Vienna, Austria, 7–12 April 2019. Anthropogenic aerosols as drivers of Mexican and Central American climate. **García-Martínez I. M.** and M. Bollasina. Poster.
- European Geosciences Union (**EGU**) General Assembly. Vienna, Austria, April 8-13, 2018. Sub-monthly variability of the Caribbean Low-Level Jet and its relationship to precipitation and atmospheric circulation in Mexico and Central America. **García-Martínez I. M.** and M. Bollasina. Poster.
- Virtual workshop of the Climate Variability and Predictability Program (**US CLIVAR**): Observing and Modelling Climate Variability in the Intra-Americas Seas and Impacts on the Continental Americas and the Caribbean. September 9-11, 2015. Ocean-atmosphere variability related with the midsummer drought in the CFSR reanalysis. **García-Martínez, I. M.** and J. Sheinbaum. Poster.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, November 2-8, 2014. Ocean-atmosphere variability related with the midsummer drought in the CFSR reanalysis. **García-Martínez, I. M.** and J. Sheinbaum. Oral presentation.
- Annual Meeting of the Mexican Geophysical Union (**UGM**). Vallarta, Mexico, October 28-November 2, 2012. Empirical model for the estimation of the hourly diffuse radiation during clear-sky conditions in Xalapa city. López-Velázquez, E. and **García-Martínez I. M.** Poster.
- **XVI Congresso Brasileiro de Meteorologia**. Belém do Pará, Brazil. September 13-17, 2010. Analysis of the minimum temperature in the 2009-2010 Winter season for the Central Mexico region. **García-Martínez, I. M.**, Gómez, G., González, P., Jofre, R., Muñoz C., and Nava, R. Oral presentation.